The Universe does Not Expand

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Abstract

Modern cosmology is based on the assumption that the redshift of galaxies is a Doppler effect resulting from the expansion of the Universe. In this paper, we show that redshift can be explained without invoking the expansion of space. We propose an alternative model in which the photon loses energy while passing through a gravitational field of density ρ. We derive the equation $z = \beta \cdot \rho_{\text{avg}} \cdot D$ and present a numerical example consistent with observational data.

The key prediction of this model is the dependence of redshift on the local mass distribution along the line of sight – distinguishing it from the homogeneous expansion paradigm. The model eliminates the need for dark energy, dark matter, and the cosmological constant.

Contents

1 Introduction

Since Edwin Hubble's discoveries in 1929, it has been assumed that the redshift of galactic spectra is caused by their recession from us, interpreted as the expansion of the Universe. However, an alternative explanation for this phenomenon has long existed – the so-called **tired light** hypothesis. This concept was rejected because it lacked a physical mechanism that would explain the linear z-distance relation.

In this work, we present a development of this concept, based on the interaction of photons with the gravitational field of real matter density ρ. We show that redshift results from the loss of energy by a photon in a gravitational field, not from a Doppler effect of receding galaxies.

The observed expansion of the Universe is an **optical illusion** resulting from the erroneous interpretation of redshift. At its core lies the assumption of the Big Bang – an event that has never been directly observed, but rather reconstructed on the basis of a flawed model. In our approach, the Universe is static, and redshift is the result of photons interacting with the gravitational field along their path from source to observer.

2 The Photon Has No Momentum

In the FBP (Photon Without Momentum) model, we have proven that a photon is not a particle with momentum $p = E/c$, but rather a **dynamic reconfiguration of the electromagnetic field**.

In the standard view, the momentum of a photon is:

$$p = \frac{E}{c}$$

Substituting $E = hc/\lambda$, we obtain:

$$p = \frac{h}{\lambda}$$

The speed of light c disappears from the equation. This means that photon momentum is a **mathematical artifact**, not a physical property. As a wave, a photon does not possess momentum in the mechanical sense – its interaction with matter is based on the work of the electric field on a charge, not on the collision of two bodies. Therefore, the two equations above have no physical meaning.

Consequently, the interaction of a photon with a gravitational field does not involve a change in momentum, but rather a change in frequency (energy) due to the work performed by the field.

3 Derivation of the Redshift Equation

Consider a photon passing through a region of constant matter density ρ over a distance D. We assume that the relative change in photon energy is proportional to the field density and the distance traveled:

$$\frac{\Delta E}{E_{\text{obs}}} = -\beta \cdot \rho \cdot D$$

where β is the coupling constant between the photon and the gravitational field.

Since the redshift is defined as:

$$z = \frac{\lambda_{\text{obs}} - \lambda_{\text{em}}}{\lambda_{\text{em}}} = \frac{E_{\text{em}} - E_{\text{obs}}}{E_{\text{obs}}}$$

for small z, we have:

$$z \approx \frac{\Delta E}{E_{\text{obs}}}$$

Thus, we obtain the fundamental equation of the model:

$$\boxed{z = \beta \cdot \rho_{\text{avg}} \cdot D}$$

3.1 Comparison with Hubble's Law

The classical Hubble law is:

$$z = \frac{H_0}{c} \cdot D$$

Comparing the two equations:

$$\frac{H_0}{c} = \beta \cdot \rho_{\text{avg}}$$

We obtain the key identity:

$$\boxed{H_0 = \beta \cdot \rho_{\text{avg}} \cdot c}$$

Thus, the Hubble constant is not a measure of expansion velocity, but the product of the coupling constant, the average matter density, and the speed of light.

4 Numerical Example

Let us take data for a galaxy at distance $D = 100\,\text{Mpc}$.

4.1 Data

- Speed of light: $c = 3 \times 10^5\,\text{km/s}$
- Measured Hubble constant: $H_0 = 67.4\,\text{km/s/Mpc}$
- Average matter density: $\rho_{\text{avg}} = 7.26 \times 10^{-27}\,\text{kg/m}^3$

- The constant β is derived from the identity:

$$\beta = \frac{H_0}{\rho_{\text{avg}} \cdot c}$$

4.2 Calculation of β

Convert H_0 to SI units:

$$H_0 = 67.4 \, \frac{\text{km/s}}{\text{Mpc}} = 67.4 \cdot \frac{10^3}{3.086 \times 10^{22}} \, \text{s}^{-1}$$

$$H_0 \approx 2.18 \times 10^{-18} \, \text{s}^{-1}$$

Now we calculate β:

$$\beta = \frac{2.18 \times 10^{-18}}{7.26 \times 10^{-27} \cdot 3 \times 10^{8}}$$

$$\beta = \frac{2.18 \times 10^{-18}}{2.178 \times 10^{-18}} \approx 1$$

We obtain $\beta \approx 1$. This means that the photon-gravitational field coupling constant is of natural magnitude – it does not require any artificial scales or additional assumptions.

4.3 Calculation of Redshift

For a distance $D = 100 \, \text{Mpc}$:

$$z = \beta \cdot \rho_{\text{avg}} \cdot D$$

We convert D to meters:

$$D = 100 \, \text{Mpc} = 100 \cdot 3.086 \times 10^{22} \, \text{m} = 3.086 \times 10^{24} \, \text{m}$$

Substituting:

$$z = 1 \cdot 7.26 \times 10^{-27} \cdot 3.086 \times 10^{24}$$

$$z \approx 7.26 \times 3.086 \times 10^{-3}$$

$$z \approx 0.0224$$

4.4 Comparison with Hubble's Law

The classical Hubble law gives:

$$z = \frac{H_0}{c} \cdot D = \frac{67.4}{3 \times 10^5} \cdot 100$$

$$z = \frac{67.4}{3000} \cdot 100 = 0.0225$$

The results are identical to within 0.4%, which is within measurement

error margins.

4.5 Conclusion

Both models yield the same value of z for $D = 100\,\mathrm{Mpc}$. The difference is conceptual:

- In the standard model: redshift = Doppler effect from receding galaxies.
- In the MRG model: redshift = energy loss of photons in the gravitational field.

5 Discussion

The presented model explains redshift without invoking the expansion of the Universe. It requires only:

- real mass (visible and dispersed invisible mass),
- a coupling constant $\beta \approx 1$,

- the speed of light c as the upper limit of field dynamics.

5.1 Key Prediction: Redshift Depends on Mass Distribution

In the expansion model (ΛCDM), redshift is solely a function of distance – space expands uniformly, so redshift is homogeneous for a given distance.

In the MRG model, redshift depends on the total gravitational field density along the photon's path:

$$z = \beta \int_0^D \rho(s)\, ds$$

This means that two photons at identical distances from Earth, but passing through different regions (e.g., a galaxy cluster versus a cosmic void), will have different redshifts. This is a testable prediction – unlike the expansion model, which offers no such possibility.

5.2 Elimination of Unnecessary Components

This model eliminates the need for:

- **Dark matter** – gravitational effects arise from dispersed mass (visible and invisible dispersed matter),
- **Dark energy** – redshift does not require accelerated expansion,
- **Cosmological constant** Λ – it is unnecessary in a static Universe.

6 Conclusion

We have presented an alternative model of redshift in which photons lose energy in a gravitational field of density ρ. We derived the equation $z = \beta\rho D$ and provided a numerical example consistent with observational data.

Key conclusions:

1. **The Universe is not expanding.** The observed redshift is the result of light energy loss in the gravitational field.
2. **Redshift depends on mass distribution** along the line of sight – a falsifiable prediction.
3. **The photon has no momentum.** Its interaction with matter is the work of the field, not a collision of particles.
4. **The Hubble constant** is a measure of the average gravitational field density: $H_0 = \beta \rho_{\text{avg}} c$.

The MRG model is simpler than the standard one, does not require unobservable components, and gives the same numerical predictions, while also delivering new, testable predictions.

The Universe is static. Light does not get tired. It loses energy. Space does not expand.

6.1 Comparison of the expansion model (ΛCDM) with the MRG model

Table 1

ΛCDM Model (expansion)	**MRG Model (static)**
Redshift = Doppler effect	Redshift = energy loss in the field
Space expands	Space is static
Requires dark matter	Does not require
Requires dark energy	Does not require
Big Bang as the beginning	No initial singularity
Redshift depends only on distance	Redshift depends on mass distribution
H_0 = expansion rate	$H_0 = \beta \rho c$

The energy lost by the photon does not disappear and is not stored in the gravitational field – the field is merely an intermediary. The field force performs work on masses, transferring the photon's energy to them, which can then be emitted as secondary radiation after mass collisions (radio waves) or dissipate as heat. In this way, the principle of conservation of energy is satisfied in a physical manner, and redshift is a natural effect of the photon's interaction with the gravitational field.

In other words: the photon's energy is dispersed among masses in the form of motion or radiation.

$$E_{\text{lost}} = \text{Work}_{\text{photon}} = \text{Work}_{\text{field}} = \Delta E_{\text{kinetic of masses}} + E_{\text{secondary radiation}}$$

This means that the photon's energy does not vanish, but is transferred to the medium.

7 The Dark Sky and the Eternal Universe – A Thought Experiment

Official physics explains the darkness of the night sky by claiming that the Universe has a finite age (the Big Bang) and that light from the most distant stars has not yet reached us, or that the source of light is receding from us due to expansion.

The explanation is simpler: the sky is dark because light from the most distant stars has simply lost energy and stretched beyond the range of our perception. A photon with a wavelength measured in kilometers is detectable by radio telescopes. To our eyes, there is a black, empty night, but in reality, an infinite, ancient cosmos pulses there.

Let us calculate this: How much time is needed for a photon to stretch to 1 km?

Let us make a quick estimate. Assume that a star emits ordinary green light (the center of the visible spectrum):

Initial wavelength:

$$\lambda_{\text{em}} \approx 500\,\text{nm} = 5 \times 10^{-7}\,\text{m}$$

Final wavelength:

$$\lambda_{\text{obs}} = 1\,\text{km} = 1000\,\text{m}$$

7.1 Calculating the redshift (z) for such a stretching

$$z = \frac{\lambda_{\text{obs}} - \lambda_{\text{em}}}{\lambda_{\text{em}}} \approx \frac{1000}{5 \times 10^{-7}} = 2 \times 10^{9}$$

The redshift must be as high as **2 billion** !
(For comparison, the most distant galaxies seen by the James Webb Space Telescope have redshifts of only about 12 to 20).

7.2 Calculating the distance (D) using the MRG formula

Given that:

$$z = \beta \cdot \rho_{\text{avg}} \cdot D$$

and with $\beta \approx 1$ and the average density of the Universe $\rho_{\text{avg}} \approx 7.26 \times 10^{-27}\,\text{kg/m}^3$, we obtain:

$$D = \frac{z}{\beta \cdot \rho_{\text{avg}}} = \frac{2 \times 10^{9}}{1 \cdot 7.26 \times 10^{-27}} \approx 2.75 \times 10^{35}\,\text{m}$$

7.3 Converting this to the photon's travel time

The time (t) required for the photon to travel at the speed of light ($c = 3 \times 10^{8}\,\text{m/s}$):

$$t = \frac{D}{c} = \frac{2.75 \times 10^{35}}{3 \times 10^{8}} \approx 9.17 \times 10^{26}\,\text{seconds}$$

A year has about 3.15×10^7 seconds, so:

$$t \approx \frac{9.17 \times 10^{26}}{3.15 \times 10^7} \approx 2.9 \times 10^{19}\,\text{years}$$

Thus, this photon would have to travel for:

$$\boxed{\mathbf{29}\ \text{trillion years!}}$$

(Three tens of trillions, i.e., 29×10^{18} years).

7.4 What does this mean ?

Official science insists that the entire Universe is only 13.8 billion years old (13.8×10^9 years).

The calculations show in black and white that if a photon from the farthest reaches of the cosmos reaches us stretched to a length of 1 km, then that photon must have originated from a galaxy billions of times farther away than the alleged "beginning of the Universe."

The Universe is not a few tens of billions of years old. It has existed for trillions (and perhaps infinitely many) years. It is stable, eternal, and gigantic.

The Big Bang theory, in light of these numbers, looks like a model that requires many invisible ingredients.

7.5 What about the Cosmic Microwave Background (CMB) ?

Astronomers observe the CMB and attribute to it a redshift of about $z \approx 1100$. They interpret this as the "echo of the Big Bang." In the MRG model, this is a natural effect of light energy loss: a photon with an initial wavelength λ_{em} is stretched about 1100 times before it reaches our detectors.

Light with a redshift of $z = 1100$ must, according to the MRG model, have traveled a distance of about 1.6×10^{13} light-years (16 trillion light-years).

This is over a thousand times more than the official age of the Universe (13.8 billion years). This means that the CMB is not an "echo of the Big Bang" – it is light that has been traveling for a long time, gradually losing energy in the gravitational field.

However, if there exist photons with $z = 10^9$ (such as in our example of green light stretched to 1 km), their energy is so low that they are below the detection threshold of our instruments. We do not register them – not because they do not exist, but because they are undetectable to us.

This explains why the night sky is dark: there is no shortage of light, we lack the sensitivity of instruments to register waves with wavelengths of kilometers. The Universe pulses with radiation invisible to us, traveling for trillions of years, giving up energy in the gravitational field.

The Universe is dark because it is old and vast.

8 The Limits of the Observable Universe – Where Light Ends and Noise Begins

8.1 The Physical Limit of Energy Loss

A photon in a gravitational field loses energy according to the equation:

$$E_{\text{obs}} = \frac{E_{\text{em}}}{1+z}$$

As distance D increases, z increases, and the photon's energy decreases. At some point, its wavelength reaches sizes comparable to the thermal noise of the environment (e.g., the CMB at 2.7 K). Then the photon becomes invisible to us – not because it ceased to exist, but because its energy is below the detection threshold.

8.2 What is CMB in the MRG Model ?

Official physics calls the CMB the "echo of the Big Bang." In the MRG model, the CMB is the cumulative radiation from the entire Universe, which has traveled so long that its waves have stretched into the millimeter and centimeter range. It is not a trace of an explosion – it is the "gravitational and thermal noise" of the very distant cosmos.

8.3 What Telescopes Do We Need ?

To look beyond the current boundary of the observable Universe (which for the Webb telescope ends in the infrared), we must move to the ultra-long wavelength bands:

- **Submillimeter and millimeter bands (1–10 mm)**
  – telescopes such as ALMA.
- **Decimeter and meter bands (10 cm – several meters)**
  – networks such as LOFAR.
- **Kilometer-wavelength band**
  – future space interferometers.

8.4 Prediction: What Will SKA Find ?

The currently built SKA (Square Kilometre Array) system will operate in the frequency range from 50 MHz to 15 GHz (wavelengths from 6 m down to centimeters).

If the MRG theory is correct, SKA should record:

1. **Galaxies and quasars with redshifts** $z > 50$ – i.e., much larger than the official model predicts.
2. **Stable structures** – i.e., structures that, according to official cosmology, "had no right to exist yet at that time."

3. **A homogeneous picture of the Universe on the longest wavelength scales** – no "dark ages" and no "epoch of reionization".

8.5 What Do We Already Know ?

The James Webb Space Telescope (JWST) is already discovering galaxies with $z \approx 15$–20, which according to the official model are "too mature" for their age. This is evidence that something is wrong with the Big Bang model.

If SKA begins to regularly map galaxies with z values in the hundreds, the official theory of "expansion from a single point" will simply collapse under its own weight.

8.6 One More Barrier – The Inverse Square Law

Even if a photon did not lose energy in the gravitational field, its intensity still decreases with the square of the distance. Light from a distant galaxy spreads over an increasingly large surface – therefore at some point it simply becomes too weak to register.

It does not matter whether it is visible light, infrared, or even gamma radiation – if the source is sufficiently distant, the number of photons reaching the detector falls below the noise threshold. The inverse square law is a physical barrier that cannot be overcome.

Therefore, what we call the "boundary of the observable Universe" is in reality the "limit of signal detectability," not the boundary of space. If we could see all the radiation, we would see an infinite, static Universe. We only see its brightest, nearest layer – and we mistake it for the entire cosmos.

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